

ARAVINDAN S

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Profile Summary

Experienced Software Engineer with expertise in AI/ML development, specializing in building and fine-tuning machine learning models, leveraging generative AI technologies and designing efficient data pipelines. Proficient in Python backend development, application development (web and desktop), full-stack technologies, and prompt engineering, delivering practical, scalable, and AI-driven solutions tailored to business needs.

Work Experience

Software Engineer

ASM Technologies Ltd - Bengaluru, Karnataka

April 2022 to Present

AI&ML Development (Dec 2022 to Present)

- Expertise in building and fine-tuning machine learning models tailored to solve specific business challenges, leveraging tools like OpenAI, Hugging Face, Google RAG, and Ollama.
- Proficient in designing data pipelines, including data preprocessing, cleaning, and transformation, ensuring high-quality inputs for model training.
- Skilled in prompt engineering, customizing and fine-tuning pre-trained models, implementing generative AI (GenAI) and LLMs to optimize performance for diverse use cases in deep learning and neural networks.
- Strong foundation in Python backend development, integrating AI models into web and desktop applications, with expertise in developing chatbots and conversational AI solutions.
- Successfully delivered multiple POCs and AI-driven tools for in-house and client projects, emphasizing scalability, innovation, and business impact.

Fullstack Development (April 2022 to Nov 2022)

- Proficient in developing web and desktop applications with a strong foundation in Python backend development using frameworks like Flask and Django.
- Skilled in frontend technologies such as React, Angular, HTML, CSS, Bootstrap, and Tailwind CSS, creating responsive and user-friendly interfaces.
- Experienced in working with databases including MongoDB, Postgres, SQL, and SQLite, ensuring efficient data management and retrieval.
- Expertise in containerization using Docker and implementing CI/CD pipelines for seamless deployment and scalable application delivery.
- Successfully contributed to full-stack projects, including a timesheet portal for internal use, showcasing end-to-end development capabilities with modern tools and technologies.

Educational Qualifications

Bachelor of Technology in Information Technology - 8.2 CGPA

Mahendra College of Engineering, Anna University- Salem, Tamil Nadu

July 2018 to June 2022

Skills

- **Programming:** Python, Java, SQL, JavaScript, HTML, CSS
- **Frameworks:** Flask, Django, React, Angular, Bootstrap, Tailwind CSS
- **Desktop Application Development:** Tkinter
- **AI/ML Libraries:**
- **Artificial Intelligence:** Transformers (Hugging Face), OpenAI API, spaCy, NLTK, Rasa
- **Computer Vision:** OpenCV, PyTorch, TensorFlow
- **Machine Learning:** NumPy, Pandas, Scikit-learn, TensorFlow, PyTorch, Keras
- **Data Visualization:** Matplotlib, Seaborn, Plotly
- **Data Science Tools:** Jupyter Notebooks, Anaconda, SciPy
- **Other Tools:** Kafka, Streamlit, ELK Stack, Jenkins
- **Cloud Platforms:** AWS, GCP
- **Version Control:** GitHub, Bitbucket
- **DevOps:** Docker, CI/CD Pipelines
- **Databases:** MongoDB, Postgres, SQLite, SQL
- **Operating Systems:** Windows, Linux (Ubuntu)

Projects

SmartPredictor App

An ASM App

June 2024 to Present

- Designed and developed a desktop application for predictive analytics using Tkinter, enabling users to upload historical data and predict new outcomes using optimized machine learning models.
- Built and integrated various machine learning models, providing performance metrics and feature importance visualization for enhanced decision-making.
- Developed robust data preprocessing pipelines, including data cleaning, transformation, and feature scaling, to ensure accurate and reliable predictions.
- Created an intuitive graphical user interface (GUI) for seamless data upload, model selection, and prediction generation.
- Implemented a secure user authentication system with login functionality to control access and enhance security.
- Optimized model accuracy and application performance to deliver real-time predictions and efficient workflows.
- Successfully automated predictive analytics workflows, simplifying complex processes for both technical and non-technical users.
- Documented the application's functionality and workflows, contributing to research and the potential publication of a research paper on its development.

AI-Powered Log Analytics Tool

Feb 2024 to May 2024

- Designed and developed a log analysis system to process and monitor live log streams using Kafka for real-time ingestion and analysis.
- Built and integrated machine learning models to classify log entries, detect anomalies, and identify errors and warnings efficiently.
- Implemented a GenAI-powered feature to provide potential resolutions for identified errors and warnings, enhancing the debugging process.
- Created an intuitive user interface using Streamlit, enabling users to visualize log data through error classifications, timestamps, error counts, and time-series plots.
- Engineered the system to display a detailed log context by analyzing lines before and after warnings/errors, ensuring precise and actionable insights.
- Enabled real-time actions like flagging, categorizing, or escalating errors directly from the interface.
- Automated workflows for continuous log ingestion and analysis, minimizing manual intervention and improving response times.
- Successfully delivered a robust tool for real-time log monitoring and error resolution, ensuring reliability and scalability for both in-house and client applications.

First Prize Winner in Code Vita for Java Programming Contest at Mahendra College of Engineering

September 2019

Awarded 1st prize in the "Code Vita for Java Programming" one-day coding contest held at Mahendra College of Engineering on 24.09.2019, showcasing exceptional coding skills and expertise in Java programming.

Certifications and Licenses

- Generative AI for Beginners, Udemy
- Introduction to AI & Machine Learning, Udemy
- ChatGPT masters
- Python Programming, Mind Luster (Credential ID: 2467375161)
- HTML, CSS, JavaScript, Mind Luster (Credential ID: 2467371123)